

Radioactivity

Worked Solutions

AF — December 2019

Introduction

This document gives worked solutions to the supplied BPhO *Radioactivity* problem sheet. The notation, constants, approximations and methods follow the sheet and its accompanying solution document. Numerical values have been evaluated explicitly so that the calculations can also be reproduced computationally.

The central model is the radioactive decay law

$$\frac{dN}{dt} = -\lambda N,$$

with solution

$$N(t) = N_0 e^{-\lambda t}.$$

The activity is

$$A(t) = -\frac{dN}{dt} = \lambda N(t),$$

and the half-life satisfies

$$\lambda = \frac{\ln 2}{t_{1/2}}.$$

Consequently,

$$\boxed{N(t) = N_0 2^{-t/t_{1/2}}} \quad \text{and} \quad \boxed{A(t) = A_0 2^{-t/t_{1/2}}}.$$

1 Question 1

1.1 (i) Half-life of radon

The radon sample falls to 2% of its original amount after 22 days:

$$\frac{N}{N_0} = 0.02.$$

Using

$$\frac{N}{N_0} = 2^{-t/t_{1/2}},$$

we obtain

$$0.02 = 2^{-22/t_{1/2}}.$$

Taking logarithms,

$$\frac{22}{t_{1/2}} = \log_2(50),$$

so

$$t_{1/2} = \frac{22}{\log_2(50)} = \boxed{3.90 \text{ days}}.$$

1.2 (ii) Carbon-14 dating

For fresh biomatter,

$$A_0 = 238 \text{ Bq kg}^{-1},$$

whereas the hammer has

$$A = 1.05 \text{ Bq kg}^{-1}.$$

Since activity follows the same exponential law as the number of nuclei,

$$\frac{A}{A_0} = 2^{-t/t_{1/2}}.$$

Therefore

$$\frac{1.05}{238} = 2^{-t/5370}.$$

Hence

$$t = 5370 \log_2 \left(\frac{238}{1.05} \right) \approx \boxed{4.20 \times 10^4 \text{ years}}.$$

Thus the hammer dates to approximately

$$\boxed{42\,000 \text{ years}}.$$

1.3 (iii) Attenuation of Sr-90 beta radiation

The intensity attenuation law is

$$I = I_0 e^{-\mu x}.$$

Here

$$I_0 = 200 \text{ Bq}, \quad I = 10 \text{ Bq}, \quad x = 2.00 \text{ mm} = 2.00 \times 10^{-3} \text{ m}.$$

Therefore

$$\frac{10}{200} = e^{-\mu(0.002)}.$$

Thus

$$\mu = \frac{\ln(20)}{0.002} \approx \boxed{1.50 \times 10^3 \text{ m}^{-1}}.$$

The half-thickness is defined by

$$\frac{I}{I_0} = \frac{1}{2},$$

so

$$\frac{1}{2} = e^{-\mu x_{1/2}},$$

giving

$$\boxed{x_{1/2} = \frac{\ln 2}{\mu}}$$

and hence

$$\boxed{x_{1/2} \approx 4.63 \times 10^{-4} \text{ m} = 0.463 \text{ mm}}.$$

1.4 (iv) Mass of Ra-226

The activity is

$$A = \lambda N,$$

so

$$N = \frac{A}{\lambda} = \frac{At_{1/2}}{\ln 2}.$$

Using

$$A = 9900 \text{ Bq}, \quad t_{1/2} = 1602 \text{ yr} \approx 1602(365.25)(24)(3600) \text{ s},$$

gives

$$N \approx 7.22 \times 10^{14} \text{ atoms}.$$

The mass of one Ra-226 atom is approximately

$$m_{\text{Ra}} = 226u, \quad u = 1.661 \times 10^{-27} \text{ kg}.$$

Hence

$$m = N(226u) \approx \boxed{2.71 \times 10^{-10} \text{ kg}}$$

or

$$\boxed{0.271 \text{ mg}}.$$

For U-235,

$$t_{1/2} = 2.22 \times 10^{16} \text{ s},$$

so

$$N_U = \frac{At_{1/2}}{\ln 2} \approx 3.17 \times 10^{20}.$$

Therefore

$$m_U = N_U(235u) \approx \boxed{1.24 \times 10^{-4} \text{ kg}} = \boxed{124 \text{ mg}}.$$

The much longer half-life of U-235 means that many more atoms are required to produce the same activity.

1.5 (v) Age of the Pa-231 statuette

The Pa-231 initially decays into Ac-227, whose subsequent decay chain eventually produces stable Pb-207. Under the problem's approximation, the measured fraction of remaining Pa-231 is therefore

$$\frac{N}{N_0} = 0.10.$$

Using

$$\frac{N}{N_0} = 2^{-t/t_{1/2}},$$

with

$$t_{1/2} = 32800 \text{ yr},$$

gives

$$0.1 = 2^{-t/32800}.$$

Thus

$$t = 32800 \log_2(10) \approx \boxed{1.09 \times 10^5 \text{ years}}.$$

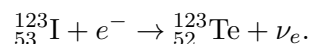
Hence the statuette has been in space for approximately

$$\boxed{109\,000 \text{ years}}.$$

2 Question 2 — Iodine-123

2.1 (i) Decay equation

Iodine has atomic number $Z = 53$. Electron capture reduces the atomic number by one without changing the mass number:



The daughter nucleus is Te-123.

2.2 (ii) Medical usefulness

I-123 has several properties that make it useful for thyroid imaging:

- its half-life of 13.22 hours is long enough for preparation and imaging, but short enough that it does not remain radioactive for an excessively long period;
- it decays by electron capture and produces a useful γ ray;
- the emitted 159 keV gamma radiation can escape the body and be detected externally;
- iodine is naturally concentrated by the thyroid, so the radioactive tracer preferentially reaches the tissue being imaged.

The radiation therefore provides a signal while allowing the distribution of iodine in the thyroid to be measured.

2.3 (iii) Activity after 24 hours

The initial activity is

$$A_0 = 2.5 \times 10^7 \text{ Bq}.$$

Using

$$A(t) = A_0 2^{-t/t_{1/2}},$$

with

$$t = 24 \text{ h}, \quad t_{1/2} = 13.22 \text{ h},$$

we obtain

$$A(24) = 2.5 \times 10^7 2^{-24/13.22}.$$

Therefore

$$\boxed{A(24) \approx 7.10 \times 10^6 \text{ Bq}}.$$

The activity actually measured in the thyroid will be lower because not all of the administered iodine is taken up and retained by the thyroid. Some activity remains elsewhere in the body and some is removed biologically.

2.4 (iv) Number of I-123 atoms

The decay constant is

$$\lambda = \frac{\ln 2}{13.22(3600)} \approx 1.456 \times 10^{-5} \text{ s}^{-1}.$$

Since

$$\begin{aligned} A_0 &= \lambda N_0, \\ N_0 &= \frac{A_0}{\lambda}. \end{aligned}$$

Hence

$$N_0 = \frac{2.5 \times 10^7}{1.456 \times 10^{-5}} \approx \boxed{1.72 \times 10^{12} \text{ atoms}}.$$

2.5 (v) Total energy from gamma emission

Each decay produces a gamma photon of energy

$$E_\gamma = 159 \text{ keV}.$$

Converting to joules,

$$E_\gamma = 159 \times 10^3 (1.602 \times 10^{-19}) \approx 2.55 \times 10^{-14} \text{ J}.$$

Assuming one such gamma emission per decay and complete absorption,

$$E_{\text{total}} = N_0 E_\gamma.$$

Thus

$$E_{\text{total}} \approx (1.72 \times 10^{12})(2.55 \times 10^{-14}) \approx \boxed{4.37 \times 10^{-2} \text{ J}}.$$

This is the total gamma energy associated with the initial population.

2.6 (vi) Initial and average gamma power

The instantaneous power immediately after administration is

$$P_0 = A_0 E_\gamma.$$

Thus

$$P_0 = (2.5 \times 10^7)(159 \times 10^3)(1.602 \times 10^{-19})$$

and therefore

$$\boxed{P_0 \approx 6.37 \times 10^{-7} \text{ W}}.$$

Over 48 hours, the exact total emitted energy is

$$E_{48} = N_0 E_\gamma \left(1 - 2^{-48/13.22}\right).$$

Hence the average power is

$$P_{\text{avg}} = \frac{E_{48}}{48(3600)} \approx \boxed{2.33 \times 10^{-7} \text{ W}}.$$

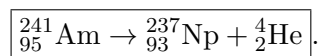
Thus the initial gamma-emission power is approximately

$$\boxed{2.7 P_{\text{avg}}}.$$

3 Question 3 — Americium-241

3.1 (i) Decay equation

Americium has atomic number 95. Alpha decay reduces the mass number by 4 and the atomic number by 2:



3.2 (ii) Activity of a fresh smoke detector

The problem states that the active material is $0.33 \mu\text{g}$ of AmO_2 . Using the supplied oxygen mass number 16, the molar mass in the problem's approximation is

$$M_{\text{AmO}_2} = 241 + 2(16) = 273 \text{ g mol}^{-1}.$$

Thus

$$n = \frac{0.33 \times 10^{-6} \text{ g}}{273 \text{ g mol}^{-1}} \approx 1.21 \times 10^{-9} \text{ mol},$$

and the number of Am-241 nuclei is

$$N_0 = nN_A \approx \boxed{7.29 \times 10^{14}}.$$

The decay constant is

$$\lambda = \frac{\ln 2}{432.2(365.25)(24)(3600)} \approx 5.08 \times 10^{-11} \text{ s}^{-1}.$$

Therefore

$$A_0 = \lambda N_0 \approx \boxed{3.70 \times 10^4 \text{ Bq}}.$$

Using

$$1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq},$$

gives

$$\boxed{A_0 \approx 1.00 \times 10^{-6} \text{ Ci}}.$$

This uses the stated AmO_2 mass directly. Treating the entire $0.33 \mu\text{g}$ as Am-241, as in the accompanying handwritten calculation, gives about $4.2 \times 10^4 \text{ Bq}$; the AmO_2 calculation is the one consistent with the wording of the question.

3.3 (iii) Ionizations and detector current

Each alpha particle has energy

$$E_\alpha = 5.486 \text{ MeV}.$$

The ionization energy is

$$E_i = 34 \text{ eV}.$$

Thus the approximate number of ion pairs produced per alpha particle is

$$n_i = \frac{5.486 \times 10^6}{34} \approx 1.61 \times 10^5.$$

Using the activity found in part (ii), the ion-pair production rate is

$$\dot{N}_i = A_0 n_i \approx (3.70 \times 10^4)(1.61 \times 10^5) \approx \boxed{5.97 \times 10^9 \text{ s}^{-1}}.$$

An ion pair produces both an electron and a positive ion. If both charge carriers are collected, their contributions to the detector current add, so

$$I = 2e\dot{N}_i.$$

Hence

$$I \approx 2(1.602 \times 10^{-19})(5.97 \times 10^9) \approx \boxed{1.91 \times 10^{-9} \text{ A}}.$$

3.4 (iv) Estimating the molar mass of smoke particles

Assume thermal equilibrium. The mean translational kinetic energy of a gas particle is

$$\frac{1}{2}M\overline{v^2} = \frac{3}{2}k_B T.$$

Thus

$$M\overline{v^2} = \text{constant}$$

at fixed temperature, so

$$M \propto \frac{1}{\overline{v^2}}.$$

The current is proportional to the rate at which the charged particles are collected. Following the model in the supplied solution, take the current reduction to correspond to a reduction of the characteristic particle speed:

$$\frac{v_{\text{smoke}}}{v_{\text{air}}} \approx 0.2.$$

Therefore

$$\frac{\overline{v_{\text{smoke}}^2}}{\overline{v_{\text{air}}^2}} \approx 0.2^2 = 0.04.$$

Hence

$$\frac{M_{\text{smoke}}}{M_{\text{air}}} = \frac{\overline{v_{\text{air}}^2}}{\overline{v_{\text{smoke}}^2}} \approx \frac{1}{0.04} = 25.$$

With

$$M_{\text{air}} = 28.97 \text{ g mol}^{-1},$$

we obtain

$$M_{\text{smoke}} \approx 25(28.97) \approx \boxed{7.24 \times 10^2 \text{ g mol}^{-1}}.$$

Thus the model predicts a characteristic smoke-particle molar mass of roughly

$$\boxed{724 \text{ g mol}^{-1}}.$$

4 Question 4 — The Elephant's Foot

4.1 (i) Absorbed particle rate

The radiation dose rate is

$$80 \text{ Gy h}^{-1} = 80 \text{ J kg}^{-1} \text{ h}^{-1}.$$

A $10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm}$ cube of water has volume

$$V = (0.1)^3 = 10^{-3} \text{ m}^3.$$

Taking the density of water to be approximately

$$1000 \text{ kg m}^{-3},$$

the mass is

$$m = 1.0 \text{ kg}.$$

Therefore the absorbed energy per second is

$$P = \frac{80 \text{ J}}{3600 \text{ s}} = 2.22 \times 10^{-2} \text{ J s}^{-1}.$$

For particle energy

$$E_p = 5 \text{ MeV} = 5 \times 10^6 (1.602 \times 10^{-19}) \approx 8.01 \times 10^{-13} \text{ J},$$

the required particle rate is

$$\dot{N} = \frac{P}{E_p} \approx \boxed{2.77 \times 10^{10} \text{ s}^{-1}}.$$

The cube is at a distance from the source. Assuming approximately isotropic spreading, the flux varies as

$$\Phi \propto \frac{1}{r^2}.$$

At three times the distance,

$$\frac{\Phi'}{\Phi} = \frac{1}{3^2} = \boxed{\frac{1}{9}}.$$

Thus the particle flux is reduced by a factor of nine.

4.2 (ii) Total activity of the Elephant's Foot

The surface area of the hemispherical cap is

$$A = \pi(a^2 + h^2),$$

with

$$a = 1.0 \text{ m}, \quad h = 0.5 \text{ m}.$$

Hence

$$A = \pi(1^2 + 0.5^2) = \boxed{3.93 \text{ m}^2}.$$

The particle rate found in part (i) is absorbed by an area

$$A_{\text{cube}} = (0.1)(0.1) = 0.01 \text{ m}^2.$$

Therefore the estimated flux is

$$\Phi = \frac{2.77 \times 10^{10}}{0.01} = 2.77 \times 10^{12} \text{ m}^{-2} \text{ s}^{-1}.$$

Multiplying by the surface area,

$$A_{\text{foot}} = \Phi A$$

gives

$$A_{\text{foot}} \approx (2.77 \times 10^{12})(3.93) \approx \boxed{1.09 \times 10^{13} \text{ Bq}}.$$

This is an order-of-magnitude estimate based on the stated assumption that the surface flux is uniform.

4.3 (iii) Lead shielding

For attenuation,

$$I = I_0 2^{-x/x_{1/2}},$$

where

$$x_{1/2} = 4.2 \text{ mm}.$$

We require

$$5 = (2.77 \times 10^8) 2^{-x/4.2 \text{ mm}}.$$

Thus

$$\frac{x}{4.2 \text{ mm}} = \log_2 \left(\frac{2.77 \times 10^8}{5} \right),$$

so

$$x = 4.2 \log_2 \left(\frac{2.77 \times 10^8}{5} \right) \text{ mm.}$$

Therefore

$$\boxed{x \approx 108 \text{ mm}}$$

or approximately

$$\boxed{10.8 \text{ cm}}.$$

5 Question 5 — Pu-238 RTG

5.1 (i) Initial atoms, activity and thermal power

The mass of plutonium is

$$m = 4.8 \text{ kg.}$$

Taking the molar mass of Pu-238 as approximately

$$M = 238 \text{ g mol}^{-1},$$

the number of moles is

$$n = \frac{4800}{238} \approx 20.17 \text{ mol.}$$

Thus

$$N_0 = nN_A \approx \boxed{1.21 \times 10^{25} \text{ atoms}}.$$

The half-life is

$$t_{1/2} = 87.7 \text{ yr,}$$

so

$$\lambda = \frac{\ln 2}{87.7(365.25)(24)(3600)} \approx 2.50 \times 10^{-10} \text{ s}^{-1}.$$

The initial activity is therefore

$$A_0 = \lambda N_0 \approx \boxed{3.04 \times 10^{15} \text{ Bq}}.$$

Each alpha particle carries

$$E_\alpha = 5.593 \text{ MeV}$$

or

$$E_\alpha = 5.593 \times 10^6 (1.602 \times 10^{-19}) \approx 8.96 \times 10^{-13} \text{ J.}$$

Assuming every alpha particle's kinetic energy becomes thermal energy,

$$P_0 = A_0 E_\alpha.$$

Hence

$$\boxed{P_0 \approx 2.73 \times 10^3 \text{ W}}.$$

The initial specific thermal power is

$$\frac{P_0}{m} = \frac{2726}{4.8} \approx \boxed{568 \text{ W kg}^{-1}}.$$

5.2 (ii) Heat power per kilogram as a function of time

The activity obeys

$$A(t) = A_0 2^{-t/t_{1/2}}.$$

Since the energy per decay is constant,

$$P(t) = P_0 2^{-t/t_{1/2}}.$$

Therefore the power per kilogram is

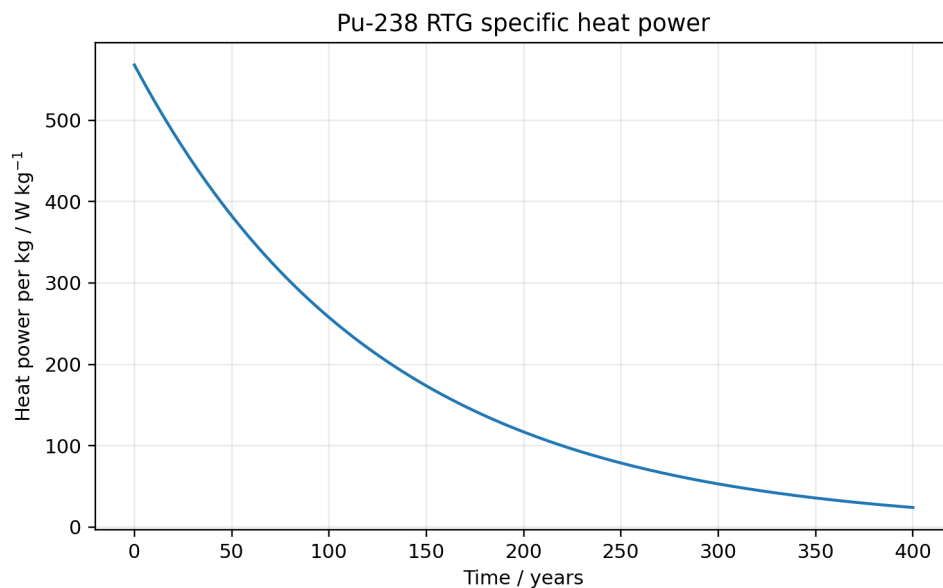
$$\boxed{\frac{P(t)}{m} = 568 2^{-t/87.7} \text{ W kg}^{-1}}.$$

The graph is therefore an exponential decay with half-life 87.7 years.

Useful reference points are

t (yr)	P/m (W kg ⁻¹)
0	568
87.7	284
175.4	142
263.1	71.0
350.8	35.5

The resulting exponential decay is plotted below.



The plotted curve uses the specific-power expression above and the reference values in the table.

6 Question 6 — Geiger–Nuttall law

6.1 (i) Constructing and testing the linear relation

The Geiger–Nuttall law supplied in the sheet is

$$\log_{10} \left(\frac{t_{1/2}}{\text{s}} \right) = a + \frac{bZ}{\sqrt{E_{\alpha}/\text{MeV}}}.$$

Define

$$x = \frac{Z}{\sqrt{E_\alpha/\text{MeV}}}, \quad y = \log_{10} \left(\frac{t_{1/2}}{\text{s}} \right).$$

The supplied spreadsheet calculates these quantities for the ten isotopes:

Isotope	Z	E_α / MeV	$\log_{10}(t_{1/2}/\text{s})$
Po-212	84	8.955	−6.52
Po-214	84	7.680	−3.80
Rn-218	86	7.263	−1.46
At-217	85	7.000	−1.49
Bi-213	83	5.870	3.44
Ra-224	88	5.789	5.50
Th-238	90	5.520	7.78
Pa-231	91	5.150	12.01
Np-237	93	4.959	13.83
Th-232	90	4.081	17.65

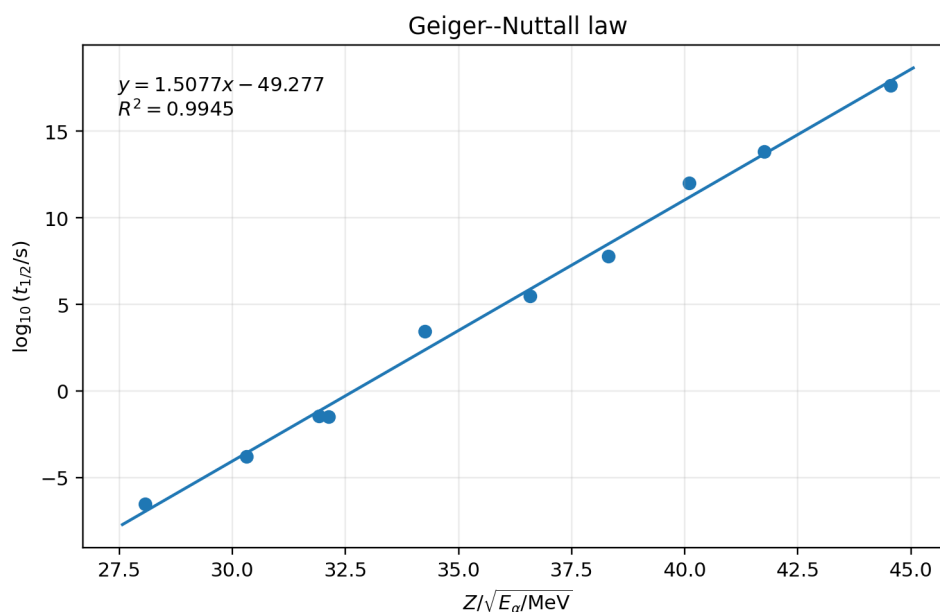
A linear regression of y against x gives

$$y = 1.5077x - 49.277$$

with

$$R^2 = 0.9945.$$

The corresponding spreadsheet-style scatter plot and fitted line are shown below.



The very high value of R^2 demonstrates that the supplied data strongly obey the linearised Geiger–Nuttall relationship.

Thus

$$\log_{10} \left(\frac{t_{1/2}}{\text{s}} \right) \approx \frac{1.51Z}{\sqrt{E_\alpha/\text{MeV}}} - 49.3.$$

6.2 (ii) Predicting the half-life of Francium-221

For Francium-221,

$$Z = 87, \quad E_\alpha = 6.3 \text{ MeV}.$$

Therefore

$$\log_{10} \left(\frac{t_{1/2}}{\text{s}} \right) \approx \frac{1.51(87)}{\sqrt{6.3}} - 49.3.$$

Numerically,

$$\frac{1.51(87)}{\sqrt{6.3}} \approx 52.34,$$

so

$$\log_{10} \left(\frac{t_{1/2}}{\text{s}} \right) \approx 3.04.$$

Hence

$$t_{1/2} \approx 10^{3.04} \text{ s} \approx \boxed{1.09 \times 10^3 \text{ s}}.$$

In minutes,

$$\boxed{t_{1/2} \approx 18.2 \text{ min}},$$

or approximately

$$\boxed{3.47 \times 10^{-5} \text{ yr}}.$$

Summary

The sheet develops a consistent computational picture of radioactive processes:

$$\boxed{\frac{dN}{dt} = -\lambda N} \quad \implies \quad \boxed{N(t) = N_0 e^{-\lambda t}}$$

with

$$\boxed{\lambda = \frac{\ln 2}{t_{1/2}}}.$$

The same exponential law governs the number of nuclei, activity and the power released by a radioactive source. Attenuation follows an analogous exponential law,

$$I = I_0 e^{-\mu x},$$

while the Geiger–Nuttall law reveals an approximately linear relationship between $\log_{10} t_{1/2}$ and $Z/\sqrt{E_\alpha}$ for alpha emitters.

These relationships provide the basis for the numerical estimates throughout the sheet, from radiocarbon dating and medical tracers to RTG power output, radiation shielding and nuclear decay timescales.